

ITEC 3100 – Introduction to Networks

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Lab 6a:

Please use a different color font for your answers so I can clearly see them:

Q1. What are the differences between private and public IP addresses?

Public is 32 bit divided into 8 bits

Private- every address is made up of a network ID and Host ID

Q2. What are the differences between IPv4 and IPv6 addresses? Why does the US still use IPv4 addressing?

IPv4- 168.28.2.98

IPv6- 10.0.52.11

It still works and it would cost a lot to switch

Q3. What are the public IP address ranges for:

i) Class A addresses: 1-126

ii) Class B addresses:127-191

iii)Class C addresses:192-223

Q4. What are the default subnet masks for:

i) Class A addresses: 255.0.0.0

- ii) Class B addresses:255.255.0.0
- iii)Class C addresses:255.255.255.0

Q5. What are the private address ranges for:

- i) Class A addresses:10.x.x.x
- ii) Class B addresses:172.16.x.x
- iii)Class C addresses:192.168.x.x

Q6. What is the purpose of private addresses and what is the APIPA address?

Private addresses are ranges used within a LAN and are not routable on public internet. APIPA is automatic private IP Addressing.

Q7. What is the NAT/PAT and why is it so important?

Network Address Translation and Port Address Translation and NAT translates private to public, PAT uses the port number to allow using a single ip address

Q8. What is special about the 127.x.x.x address range and what is 127.0.0.1 used for? Loopback address for testing and reserved

Q9. If you get a 169.254.x.x address what is most likely the issue? **Device cannot connect to the network to obtain a valid IP address.**

Q10. What is the purpose of the **default gateway** and what device does it usually represent? **Used in IP config and usually the address of a router interface.**

Q11. Run the command **netstat**, insert screenshot and describe what you are seeing. **List of my active network connections**

```
Address: 23.53.11.171
[shill@mac ~ % netstat
Active Internet connections
Proto Recv-Q Send-Q Local Address Foreign Address (state)
tcp4 0 0 mac.lan.56062 52.111.230.6.https ESTABLISHED
tcp4 0 0 mac.lan.56028 yx-in-f100.1e100.ima ESTABLISHED
tcp4 0 0 macbook-air.local.56021 ashilles-iphone.49531 ESTABLISHED
tcp4 0 0 mac.lan.56027 yx-in-f100.1e100.ima ESTABLISHED
tcp4 0 0 mac.lan.56026 yx-in-f100.1e100.ima ESTABLISHED
tcp4 0 0 macbook-air.local.black fe80::fbd3:3aba:1026 ESTABLISHED
tcp4 0 0 macbook-air.local.1026 fe80::fbd3:3aba:1026 ESTABLISHED
tcp4 0 0 mac.lan.56049 52.118.1.38.443 TIME_WAIT
tcp4 0 0 mac.lan.56048 52.182.143.215.443 ESTABLISHED
tcp4 0 0 mac.lan.56442 17.57.144.155.5223 ESTABLISHED
udp4 0 0 *.50602 *.*
udp4 0 0 *.dhcpv6-server *.*
udp4 0 0 *.domain *.*
udp4 0 0 *.domain *.*
udp4 0 0 *.bootps *.*
udp4 0 0 *.53842 *.*
udp4 0 0 *.55137 *.*
udp4 0 0 *.*
udp4 0 0 *.*
udp4 0 0 *.*
udp4 0 0 *.*
udp4 0 0 *.*
udp4 0 0 *.*
udp4 0 0 *.61132 *.*
udp4 0 0 *.*
udp4 0 0 *.*
udp4 0 0 *.*
udp4 0 0 *.*
udp4 0 0 *.*
udp4 0 0 *.*
udp4 0 0 *.*
udp4 0 0 *.*
udp4 0 0 *.mdns *.*
udp4 0 0 *.mdns *.*
udp4 0 0 *.netbios-ns *.*
udp4 0 0 *.netbios-dgm *.*
Active Multipath Internet connections
Proto/ID Flags Local Address Foreign Address (state)
icmp 0 0 *.*
icmp 0 0 *.*
Active LOCAL (UNIX) domain sockets
Address Type Recv-Q Send-Q Inode Conn Refs Nextref Addr
29d6a8c2e0f0a37 stream 0 0 0 d4f49fea3918e92 0 0 /private/var/run/cupsd
d4f49fea3918e92 stream 0 0 0 29d6a8c2e0f0a37 0 0
7c57d94f0c7b6351 stream 0 0 0 8b4643c1cc2fd01 0 0 /var/run/mDNSResponder
8b4643c1cc2fd01 stream 0 0 0 7c57d94f0c7b6351 0 0
3a8dc274832e9d38 stream 0 0 0 7128ed1fff5bb355 0 0 /var/run/mDNSResponder
7128ed1fff5bb355 stream 0 0 0 3a8dc274832e9d38 0 0
e210fe4b4fa4317c stream 0 0 0 8d4e29de02999309 0 0 /var/run/mDNSResponder
```

Q12. How are those port numbers represented and selected? **They are represented by name and by number, they are selected by range.**

Q13. Run the command **nslookup** www.wellsfargo.com , insert screenshot and describe what you see. I see the server and Ip address, I also see multiple names and addresses

```
Last login: Tue Mar 17 09:28:56 on ttys000
shill@mac ~ % nslookup www.wellsfargo.com
Server:      192.168.86.1
Address:     192.168.86.1#53

Non-authoritative answer:
www.wellsfargo.com canonical name = www.wellsfargo.com.edgekey.net.
www.wellsfargo.com.edgekey.net canonical name = e82164.b.akamaiedge.net.
Name:   e82164.b.akamaiedge.net
Address: 23.53.11.174
Name:   e82164.b.akamaiedge.net
Address: 23.53.11.172
Name:   e82164.b.akamaiedge.net
Address: 23.53.11.175
Name:   e82164.b.akamaiedge.net
Address: 23.53.11.171

shill@mac ~ %
```

Q14. From your home laptop type in: what is my ip address into google and insert screenshot.

whatismyipaddress.com

Enter Keywords or IP Address... Search

ABOUT PRESS PODCAST SUPPORT

MY IP IP LOOKUP HIDE MY IP VPNs TOOLS LEARN

My IP Address is:

IPv6: **2a04:4e41:7003:bed1::c8f6:9ed1**

IPv4: **146.75.248.147**

My IP Information:

ISP: iCloud Private Relay

Services: [Apple iCloud Private Relay](#)

City: Lawrenceville

Region: Georgia

Country: United States

Looks like you're using a VPN!

RATE YOUR VPN

[Show Complete IP Details](#)

Chattanooga

Click for more details about **2a04:4e41:7003:bed1::c8f6:9ed1**

Atlanta Athens

Location Update

Ad Switch to Mint Mobile

Shop Now

Which is your best... ng the Internet?

Q15. From your cell phone at home type in: what is my ip address into google and insert screenshot. Is this the same IP as in Q14? **No**

8:01

94



whatismyipaddress.com



WhatIsMyIPAddress.com



Enter Number & Search For Name

Enter Phone Number

EX: (888) 555-0100

BeenVerified

SEARCH

My IP Address is:

IPv4: ? 96.32.64.202

IPv6: ? Not detected

Chattanooga

Greenville

erson

Gadsden

Click for more details about 96.32.64.202

Your out-of-office days

Skip Ad

Lenovo

Enter Number & Search For Name

Enter Phone Number

EX: (888) 555-0100

BeenVerified

SEARCH



Complete and upload to the Lab 6 dropbox.